Alexandria N. Hughes, Ph.D.

hughes.alexnicole@gmail.com
https://hughes.science

Education & Training

University of Colorado, Anschutz Medical Campus — Aurora, Colorado
PhD, Neuroscience — Aug 2015 – Sep 2020

- Mentor: Bruce Appel
- Dissertation: Neuronal and microglial interactions with oligodendrocytes regulate myelin sheath formation and elimination

Texas A&M University — College Station, Texas
MS, Statistics — Jan 2021 – current

Boise State University — Boise, Idaho
BS, Biology, Mathematics minor — Aug 2010 – Dec 2014

Funding

- **National Science Foundation Graduate Research Fellowship Program** — 2016 – 2019
- **Hilda D. Elliott University Foundation Research Fellowship** — 2014 – 2015
- **Boise State University Student Research Initiative** — Summer 2014

Publications

- **Hughes AN** & Appel B (2020). Microglia phagocytose myelin sheaths to modify developmental myelination. *Nature Neuroscience*, 23, 1055–1066.
- **Hughes AN** & Appel B (2019). Oligodendrocytes express synaptic proteins that modulate myelin sheath formation. *Nature Communications*, 10(1), 1-15.
- **Hughes AN**, Oxford A, Tawara K, Jorcyk C, Oxford JT (2017). Endoplasmic reticulum stress and unfolded protein response in cartilage pathophysiology; contributing factors to apoptosis and osteoarthritis. *International Journal of Molecular Sciences*, 18(3), 665.
- **Hughes AN**, Hixon AM, Josey M (2016). A Schwanncentric View of Axon Arborization in Neuromuscular Junction (NMJ) Formation. *Journal of Neuroscience*, 36(38), 9760-9762.
- **Hughes AN** & Oxford JT (2014). A lipid-rich gestational diet predisposes offspring to nonalcoholic fatty liver disease: a potential sequence of events. *Hepatic Medicine*, 6, 15-23.

Talks & Seminars

- Open Box Science, Neuroimmunology & Neurodevelopment (Virtual) — 8/21/2020
- CSHL Zebrafish Neural Circuits & Behavior, Cold Spring Harbor, NY — 11/20/2019
- Neuroscience Graduate Program Annual Retreat, Keystone, CO — 11/15/2019
- Seminar: NYU Langone, Neuroscience Institute, Host: David Schoppik — 10/31/2019
- Seminar: Regis University, Dept of Biology Seminar Series, Host: Karin Streifel — 9/30/2019
- Rocky Mountain Regional Neuroscience Group Annual Meeting, Aurora, CO — 4/26/2019
- Gordon Research Seminar: Glial Biology, Ventura, CA — 3/3/2019
- CSHL Glia in Health & Disease, Cold Spring Harbor, NY — 7/21/2018

Poster Presentations

- CSHL Glia in Health & Disease (Virtual) — 7/21/2020
- Gordon Research Conference: Glial Biology 2019, Ventura, CA — 3/4/2019

- Neuroscience Graduate Program Annual Meeting, Estes Park, CO *best poster award 11/9/2018
- Society for Neuroscience 2017, Washington, DC 11/13/2017
- Biomolecular Research Center Open House, Boise, ID 12/8/2014
- Boise State University Undergraduate Research Conference 2014, Boise, ID 4/21/2014

Teaching

- Adjunct faculty, Dept of Mathematics, Boise State University 2014 – 2015
  - Elementary Algebra (MATH 025), 2 sections
- Tutor, Math Drop-in Center, Dept of Mathematics, Boise State University 2011 – 2014
  - Drop-in tutoring (30 hr/wk) for all lower-division mathematics courses
- Learning Assistant, Advising & Academic Enhancement, Boise State University Fall 2013
  - Calculus II (MATH 175), William (Billy) Hudson, 2 sections
- Teaching Assistant, Dept of Mathematics, Boise State University 2012 – 2013
  - Multivariable & Vector Calculus (MATH 275), Dr. Shari Ultman, 3 sections

Service

- Admissions committee student member, CU Neuroscience Graduate Program 2016 – 2019
- Student organizer, RMRNG (local SFN) annual meeting 2018 2017 – 2018
  - "Glial contributions to circuit function" May 11, 2018
  - Invited and hosted Drs. Marc Freeman, Dori Schafer, & Gabriel Corfas
- CU Neuroscience first-year bootcamp instructor, Molecular Cloning 2016 – 2018

Awards

- Werner & Kitty Hirs Travel Award, CSHL Zebrafish Neural Circuits & Behavior 2019
- Best Poster Award, Neuroscience Graduate Program annual retreat 2018
- Werner & Kitty Hirs Travel Award, CSHL Glia in Health & Disease 2018
- **National Science Foundation Graduate Research Fellowship Program** 2016 – 2019
- Victoria Croft Biology Award 2014 – 2015
- Darin Fralick Biology Award 2014 – 2015
- BSU Biomolecular Sciences PhD Program Undergraduate Grant & Internship Spring 2014
- BSU Mathematics Endowed Scholarship 2013 – 2014
- BSU Department of Mathematics Scholarship 2013 – 2014
- BSU Department of Biological Sciences Scholarship 2013 – 2014
- BSU Department of Biological Sciences Scholarship 2011 – 2012